DO NOW — RECALL

Symmetric and asymmetric encryption

Name: _____ Class: _____ Date: _____

1. Which technique protects data against casual interception?

2. What does a network proxy do?

3. What is the difference between confidentiality and integrity?

4. Rising-edge D flip-flop Q=0, D=1 just before next edge. What happens to Q, and when?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Encryption.

2. Model answer

Acts as an intermediary for requests/responses.

3. Model answer

Confidentiality prevents unauthorised reading; integrity detects unauthorised changes.

4. Model answer

Q changes to 1 on the next rising edge, not simply when D changes.

Vocabulary — know these terms

symmetric — Shared-secret encryption.

asymmetric — Public/private-key cryptography.

public key — Distributable key in a pair.

private key — Secret member of key pair.

TLS — Protocol establishing secure authenticated channels.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Show method

Bob retains the private key and decrypts. The public key must be authenticated to prevent substitution.

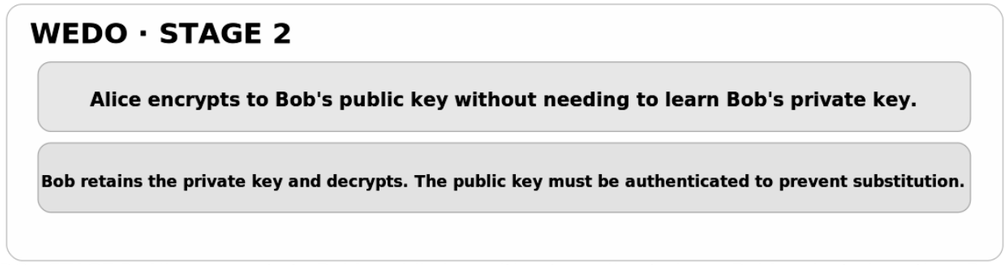
IMPROVE: Explain one correction to your own reasoning.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Bob posts his public encryption key. Alice wants only Bob to read a message. Which key is used?

Alice encrypts to Bob's public key without needing to learn Bob's private key.



WEDO · STAGE 2: Alice encrypts to Bob's public key without needing to learn Bob's private key. then Bob retains the private key and decrypts. The public key must be authenticated to prevent substitution.

NEW LEARNING

Read, annotate and explain

Understand

Symmetric encryption uses the same secret key for encryption and decryption. The difficulty is sharing that key securely.

Trace

Asymmetric encryption uses a public/private key pair. Public-key encryption for confidentiality uses a recipient's public key to encrypt and their private key to decrypt.

Evaluate

Real secure connections typically use a hybrid protocol: authenticated asymmetric cryptography to establish secrets, then efficient symmetric encryption of application data.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

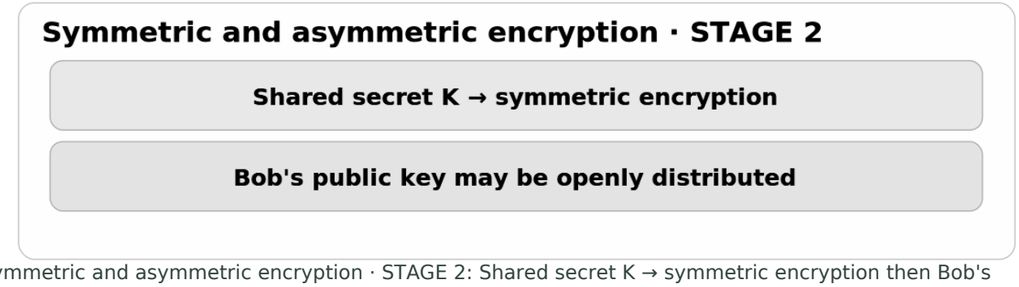
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



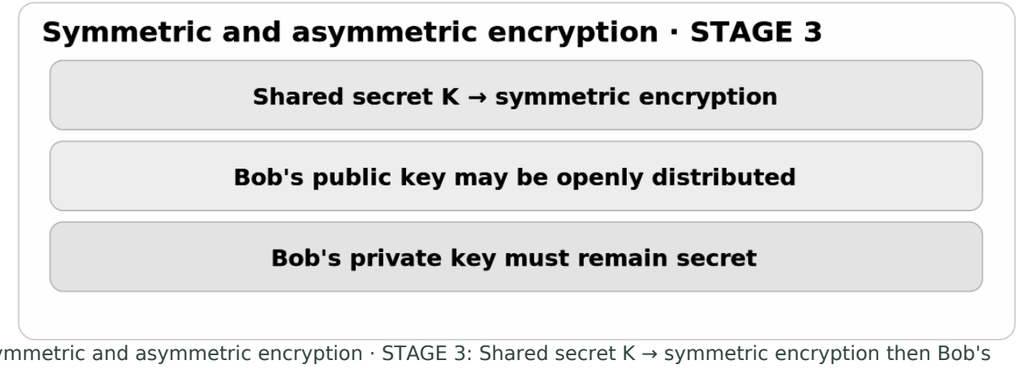
Symmetric and asymmetric encryption · STAGE 1: Shared secret K → symmetric encryption

New Learning



Symmetric and asymmetric encryption · STAGE 2: Shared secret K → symmetric encryption then Bob's public key may be openly distributed

New Learning



Symmetric and asymmetric encryption · STAGE 3: Shared secret K → symmetric encryption then Bob's public key may be openly distributed then Bob's private key must remain secret

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A public key must be kept secret to use public-key encryption.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

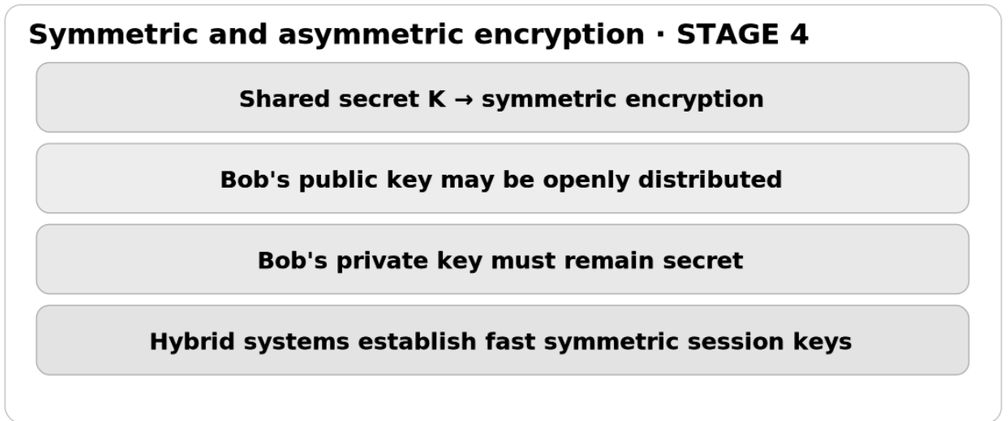
Question 3

Evaluate symmetric-only encryption for a school communicating with 200 external partners. [6]

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Symmetric and asymmetric encryption · STAGE 4: Shared secret K → symmetric encryption then Bob's public key may be openly distributed then Bob's private key must remain secret then Hybrid systems establish fast symmetric session keys

I DO — WORKED EXAMPLE

Follow the worked method

Alice and Bob already share the secret key K. Which key does each use to protect and recover a file?

Predict first

Alice encrypts with the agreed secret K; an observer without K should not recover the plaintext.

Show method

Bob uses the same secret K to decrypt, so key secrecy and its secure distribution matter.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Describe symmetric and asymmetric key arrangements. [6]

Question 2

Explain how a secure website can combine asymmetric and symmetric methods. [6]

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first Alice encrypts to Bob's public key without needing to learn Bob's private key.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Alice and Bob already share the secret key K. Which key does each use to protect and recover a file?

Alice encrypts with the agreed secret K; an observer without K should not recover the plaintext.

IDO · STAGE 1

Alice encrypts with the agreed secret K; an observer without K should not recover the plaintext.

IDO · STAGE 1: Alice encrypts with the agreed secret K; an observer without K should not recover the plaintext.

Alice and Bob already share the secret key K. Which key does each use to protect and recover a file?

Alice encrypts with the agreed secret K; an observer without K should not recover the plaintext.

IDO · STAGE 2

Alice encrypts with the agreed secret K; an observer without K should not recover the plaintext.

Bob uses the same secret K to decrypt, so key secrecy and its secure distribution matter.

IDO · STAGE 2: Alice encrypts with the agreed secret K; an observer without K should not recover the plaintext. then Bob uses the same secret K to decrypt, so key secrecy and its secure distribution matter.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Bob posts his public encryption key. Alice wants only Bob to read a message. Which key is used?

Alice encrypts to Bob's public key without needing to learn Bob's private key.



WEDO · STAGE 1: Alice encrypts to Bob's public key without needing to learn Bob's private key.